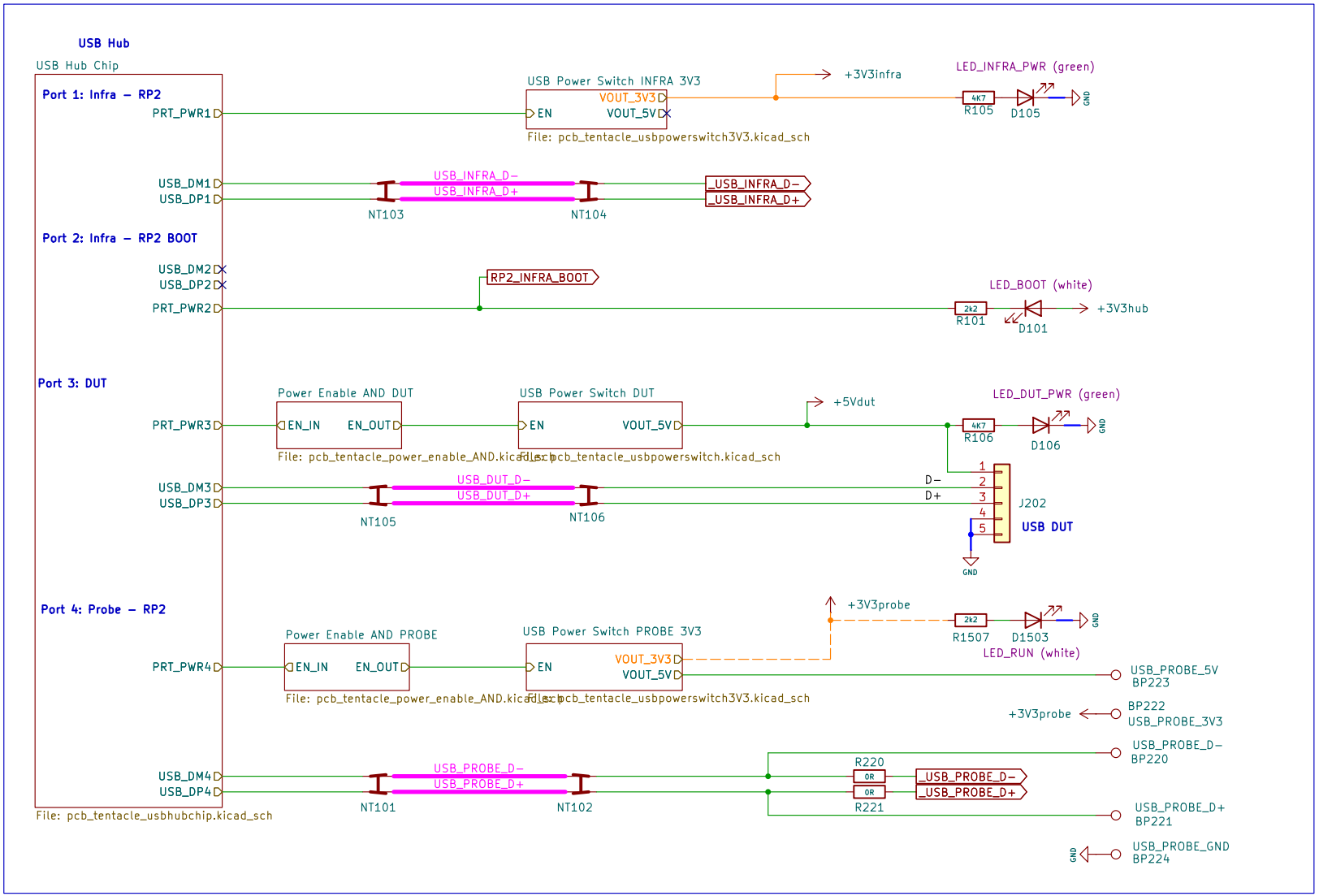


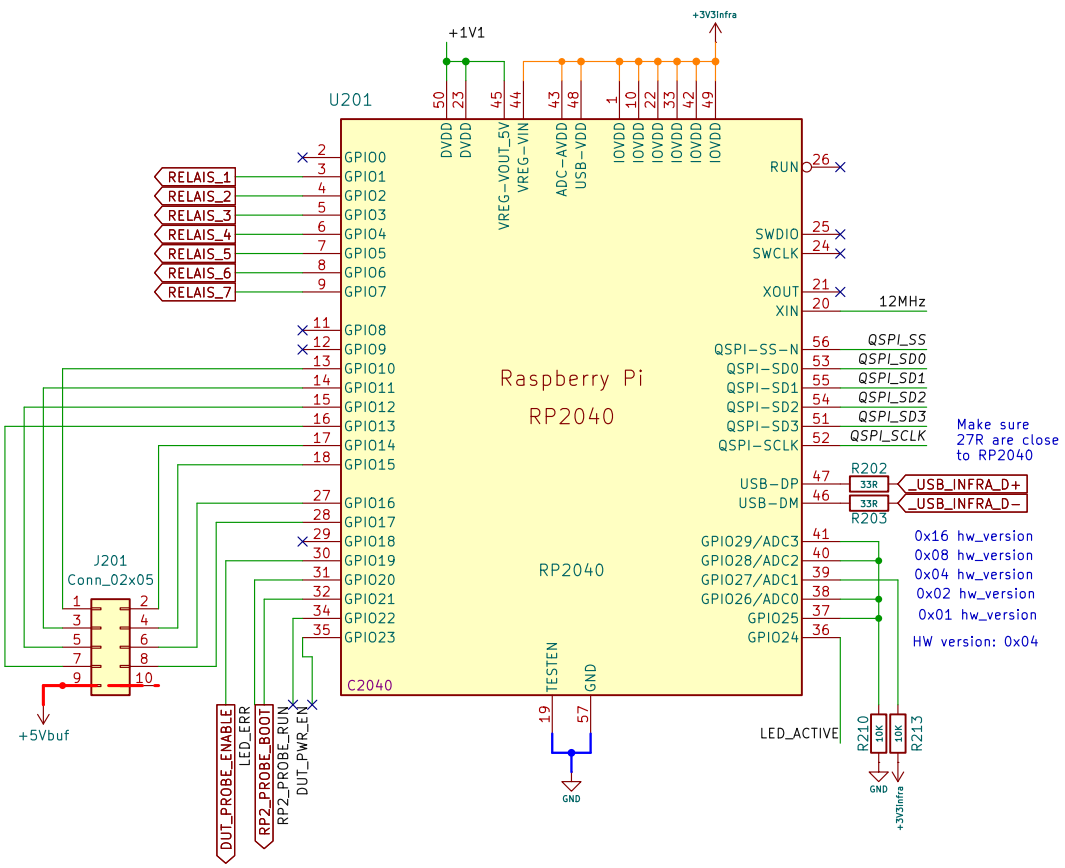
The MIT License (MIT)
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Sheet: /
File: pcb_tentacle.kicad_sch

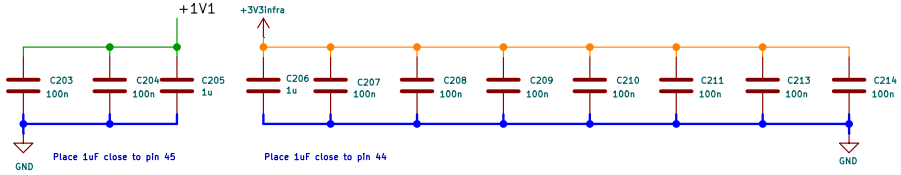
Title: Octoprobe tentacle

Size: A4	Date: 2026-10-12	**Rev: 0.9.0**
KiCad E.D.A. 10.0.4		Id: 1/21

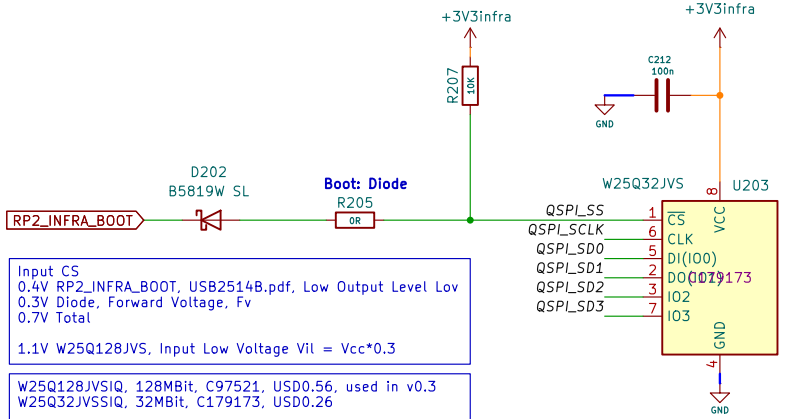
RP2 INFRA

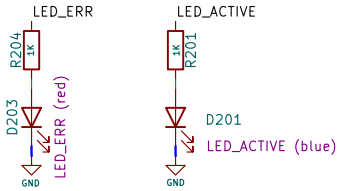


Bypass Capacitors

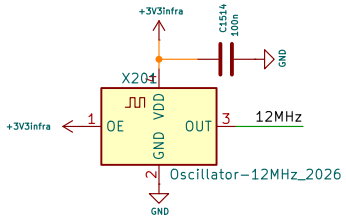


Flash Memory





Oscillator

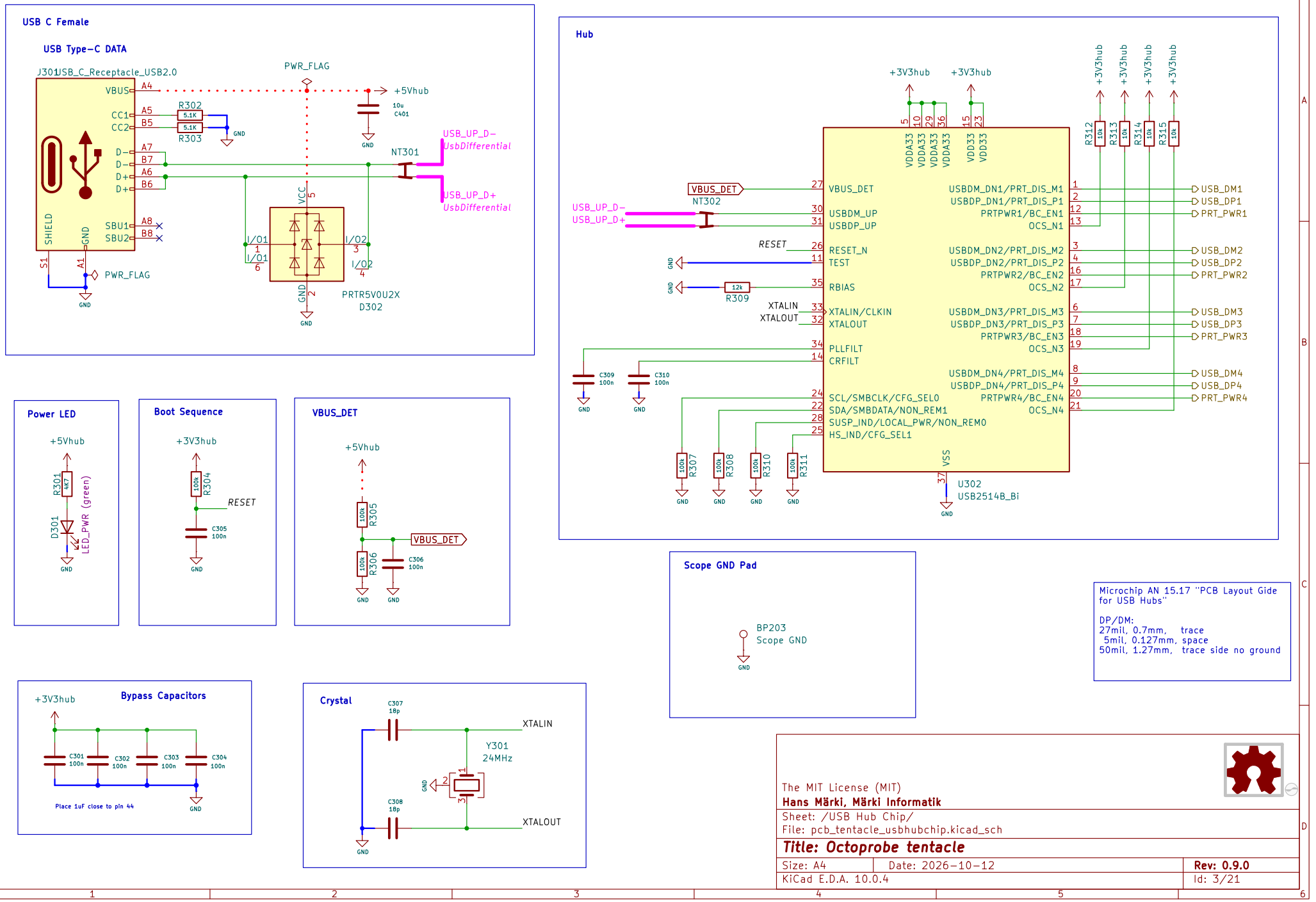


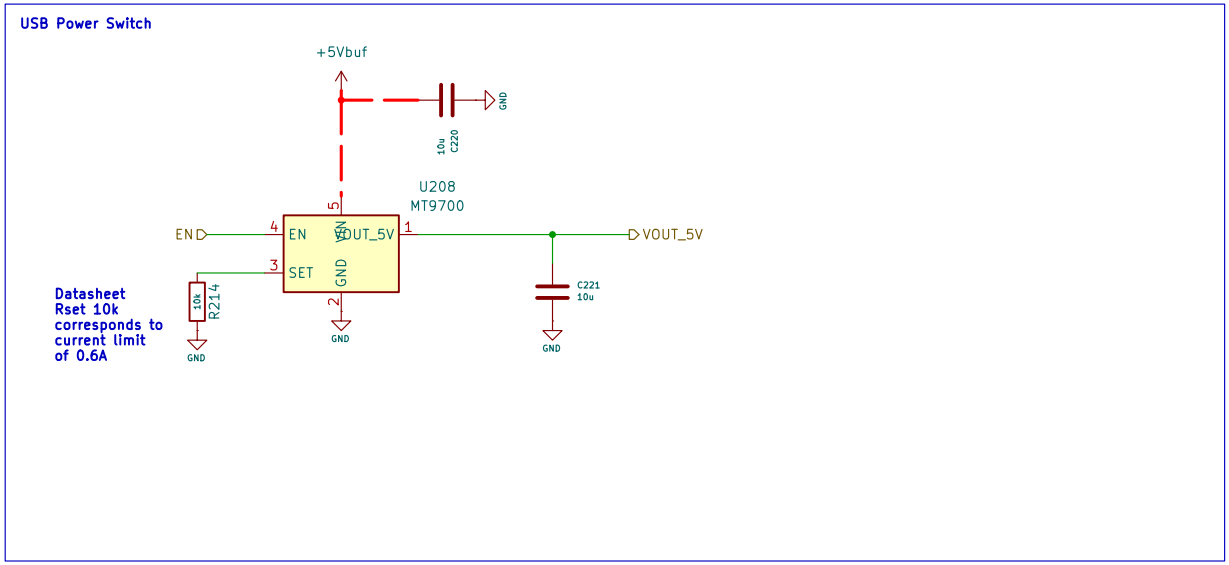
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Sheet: /RP2 INFRA/
File: pcb_tentacle_rp2_infra.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2026-10-12	**Rev: 0.9.0**
KiCad E.D.A. 10.0.4		Id: 2/21


USB C Female
USB Type-C DATA
J301USB_C_Receptacle_USB2.0
PWR_FLAG
+5Vhub
R302
R303
5.1K
10u
C401
GND
USB_UP_D-
UsbDifferential
USB_UP_D+
NT301
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
S1
A1
A4
A5
B5
A7
B7
A6
B6
A8
B8
VCC
I/O1
I/O2
PRTR5V0U2X
D302
Hub
+3V3hub
VDDA33
VDD33
VBUS_DET
NT302
USB_UP_D-
USB_UP_D+
RESET
R309
12k
XTALIN
XTALOUT
C309
C310
100n
VBUS_DET
USBDM_UP
USBDP_UP
RESET_N
TEST
RBIAS
XTALIN/CLKIN
XTALOUT
PLLFILT
CRFILT
SCL/SMBCLK/CFG_SEL0
SDA/SMBDATA/NON_REM1
SUSP_IND/LOCAL_PWR/NON_REM0
HS_IND/CFG_SEL1
VSS
U302
USB2514B_Bi
USBDM_DN1/PRT_DIS_M1
USBDP_DN1/PRT_DIS_P1
PRTPWR1/BC_EN1
OCS_N1
USBDM_DN2/PRT_DIS_M2
USBDP_DN2/PRT_DIS_P2
PRTPWR2/BC_EN2
OCS_N2
USBDM_DN3/PRT_DIS_M3
USBDP_DN3/PRT_DIS_P3
PRTPWR3/BC_EN3
OCS_N3
USBDM_DN4/PRT_DIS_M4
USBDP_DN4/PRT_DIS_P4
PRTPWR4/BC_EN4
OCS_N4
R307
R308
R310
R311
R312
R313
R314
R315
10k
USB_DM1
USB_DP1
PRT_PWR1
USB_DM2
USB_DP2
PRT_PWR2
USB_DM3
USB_DP3
PRT_PWR3
USB_DM4
USB_DP4
PRT_PWR4
Power LED
+5Vhub
R301
4K7
D301
LED_PWR (green)
Boot Sequence
+3V3hub
R304
100k
RESET
C305
100n
VBUS_DET
+5Vhub
R305
R306
C306
Scope GND Pad
BP203
Scope GND
Microchip AN 15.17 "PCB Layout Gide for USB Hubs"
DP/DM:
27mil, 0.7mm, trace
5mil, 0.127mm, space
50mil, 1.27mm, trace side no ground
Bypass Capacitors
+3V3hub
C301
C302
C303
C304
100n
Place 1uF close to pin 44
Crystal
C307
C308
18p
XTALIN
XTALOUT
Y301
24MHz
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Sheet: /USB Hub Chip/
File: pcb_tentacle_usbhubchip.kicad_sch
Title: Octoprobe tentacle
Size: A4
Date: 2026-10-12
Rev: 0.9.0
KiCad E.D.A. 10.0.4
Id: 3/21

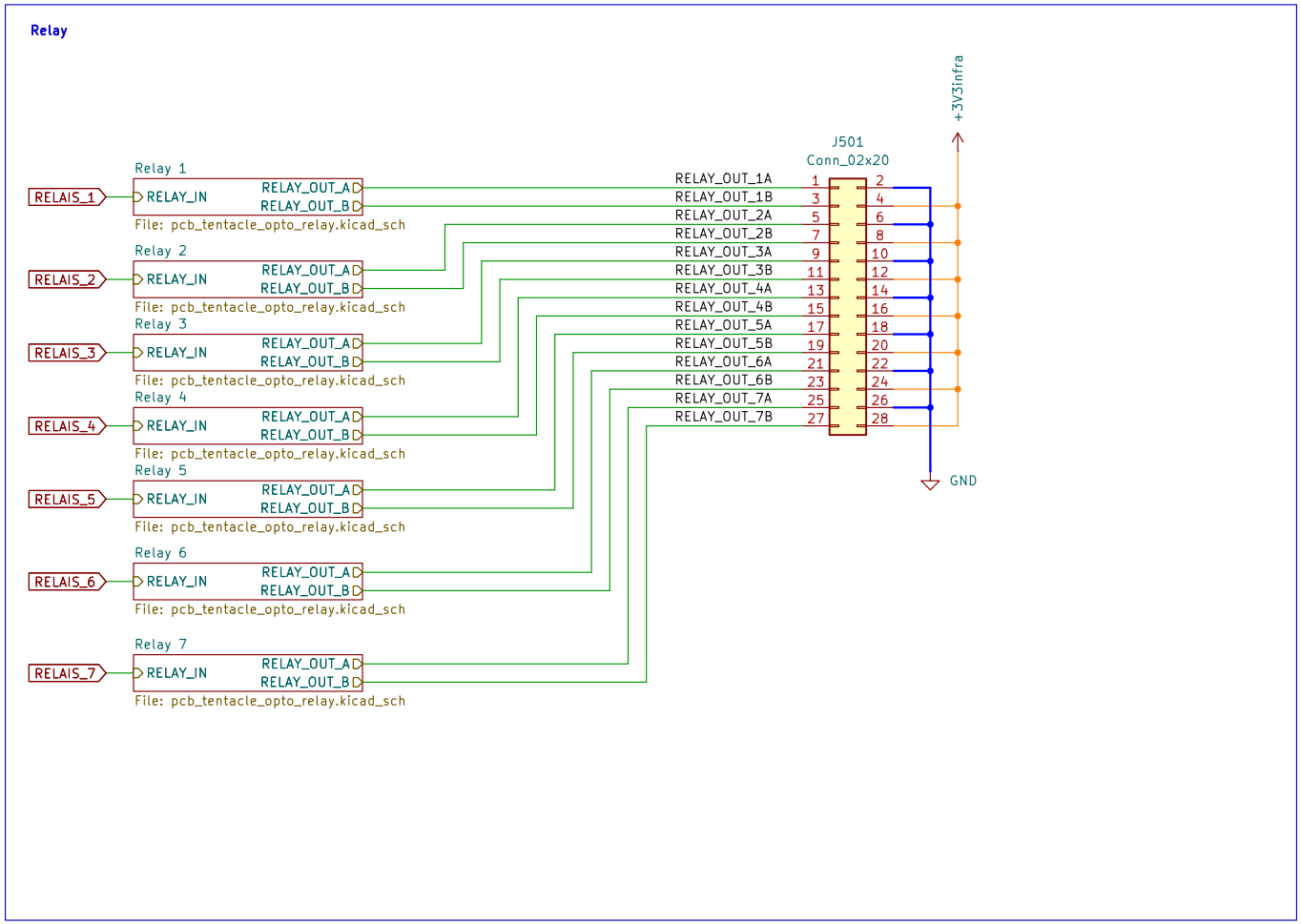


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Sheet: /USB Power Switch DUT/
File: pcb_tentacle_usbpowerswitch.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2025-02-13	**Rev: 0.4**
KiCad E.D.A. 10.0.4		Id: 4/21

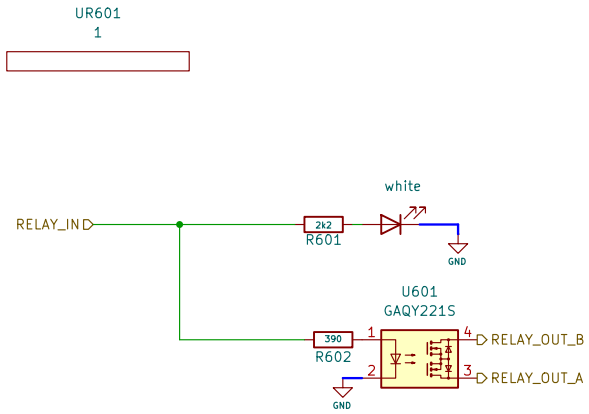


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Sheet: /Relay Breakout/
File: pcb_tentacle_relay_breakout.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2026-10-12	**Rev: 0.9.0**
KiCad E.D.A. 10.0.4		Id: 5/21



Calculation of the input resistor

Vdiff = 3.3V – Vf = 3.3V– 1.2V = 2.1V
I_Fon = 5mA
R = 2.1V / 5mA = 420 Ohm

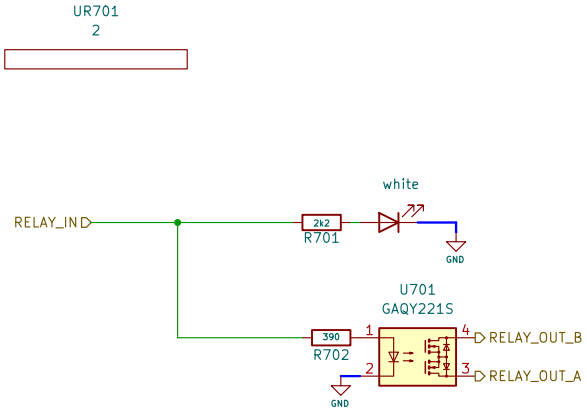
390 Ohm

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Sheet: /Relay Breakout/Relay 1/
File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2026-10-12	**Rev: 0.9.0**
KiCad E.D.A. 10.0.4		Id: 6/21



Calculation of the input resistor

Vdiff = 3.3V – Vf = 3.3V– 1.2V = 2.1V
I_Fon = 5mA
R = 2.1V / 5mA = 420 Ohm

390 Ohm

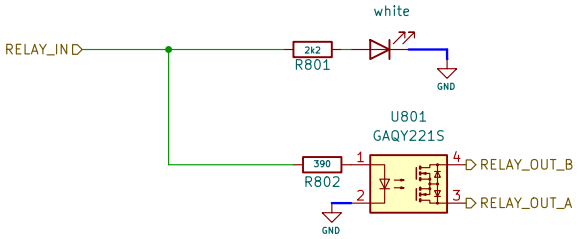
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Sheet: /Relay Breakout/Relay 2/
File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2026-10-12	**Rev: 0.9.0**
KiCad E.D.A. 10.0.4		Id: 7/21

UR801
3



Calculation of the input resistor

Vdiff = 3.3V – Vf = 3.3V– 1.2V = 2.1V
I_Fon = 5mA
R = 2.1V / 5mA = 420 Ohm

390 Ohm

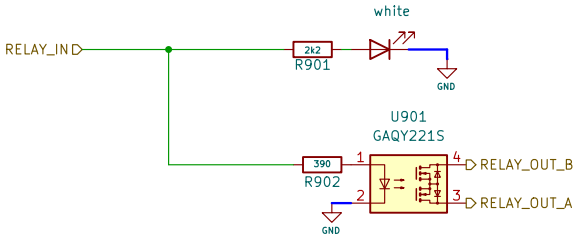
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Sheet: /Relay Breakout/Relay 3/
File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2026-10-12	**Rev: 0.9.0**
KiCad E.D.A. 10.0.4		Id: 8/21

UR901
4



Calculation of the input resistor

Vdiff = 3.3V – Vf = 3.3V– 1.2V = 2.1V
I_Fon = 5mA
R = 2.1V / 5mA = 420 Ohm

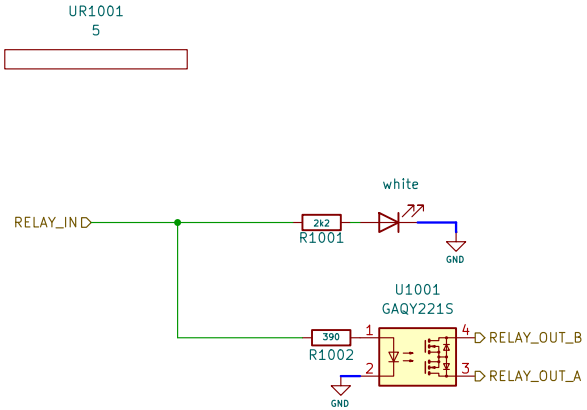
390 Ohm

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Sheet: /Relay Breakout/Relay 4/
File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2026-10-12	**Rev: 0.9.0**
KiCad E.D.A. 10.0.4		Id: 9/21



Calculation of the input resistor

Vdiff = 3.3V – Vf = 3.3V– 1.2V = 2.1V
I_Fon = 5mA
R = 2.1V / 5mA = 420 Ohm

390 Ohm

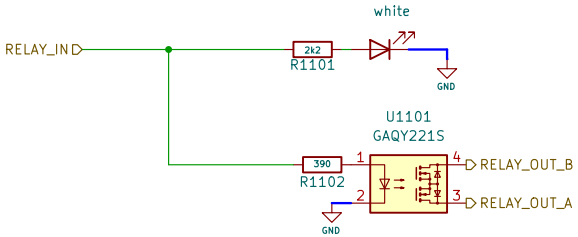
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Sheet: /Relay Breakout/Relay 5/
File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2026-10-12	**Rev: 0.9.0**
KiCad E.D.A. 10.0.4		Id: 10/21

UR1101
6



Calculation of the input resistor

Vdiff = 3.3V – Vf = 3.3V– 1.2V = 2.1V
I_Fon = 5mA
R = 2.1V / 5mA = 420 Ohm

390 Ohm

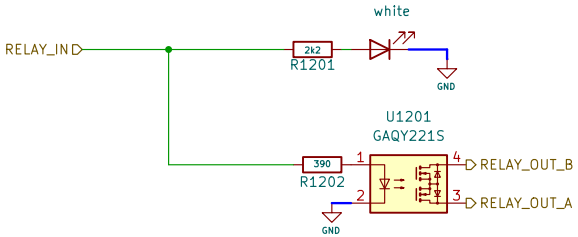
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Sheet: /Relay Breakout/Relay 6/
File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2026-10-12	**Rev: 0.9.0**
KiCad E.D.A. 10.0.4		Id: 11/21

UR1201
7



Calculation of the input resistor

Vdiff = 3.3V – Vf = 3.3V– 1.2V = 2.1V
I_Fon = 5mA
R = 2.1V / 5mA = 420 Ohm

390 Ohm

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Sheet: /Relay Breakout/Relay 7/
File: pcb_tentacle_opto_relay.kicad_sch

Title: Octoprobe tentacle

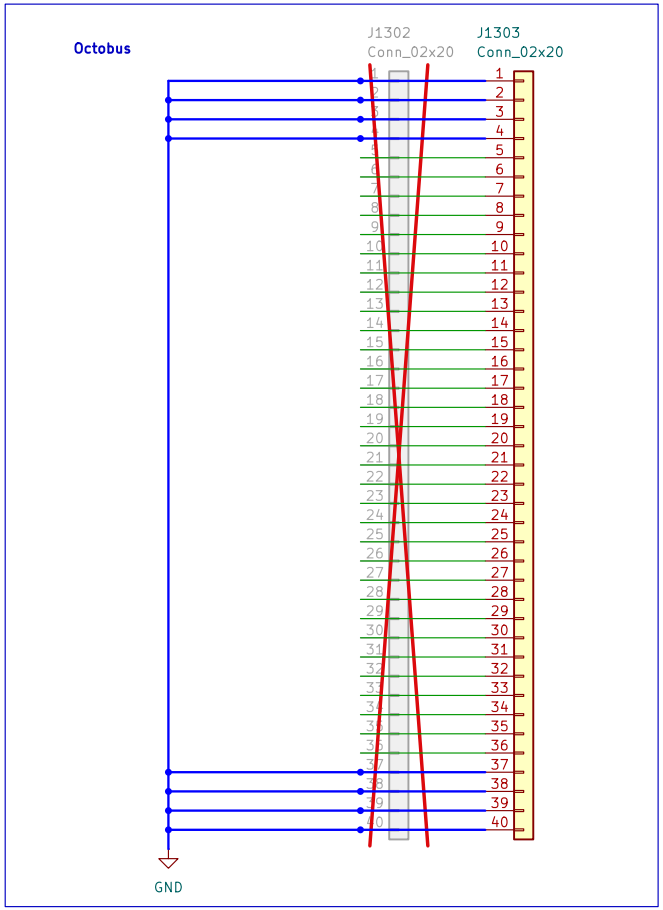
Size: A4	Date: 2026-10-12	**Rev: 0.9.0**
KiCad E.D.A. 10.0.4		Id: 12/21

Prototype Area

H1305
octoprobe_MountingHole
H1306
octoprobe_MountingHole
H1307
octoprobe_MountingHole
H1308
octoprobe_MountingHole
GND

TH1301
JLCPCB Tooling Hole
TH1302
JLCPCB Tooling Hole

Octobus

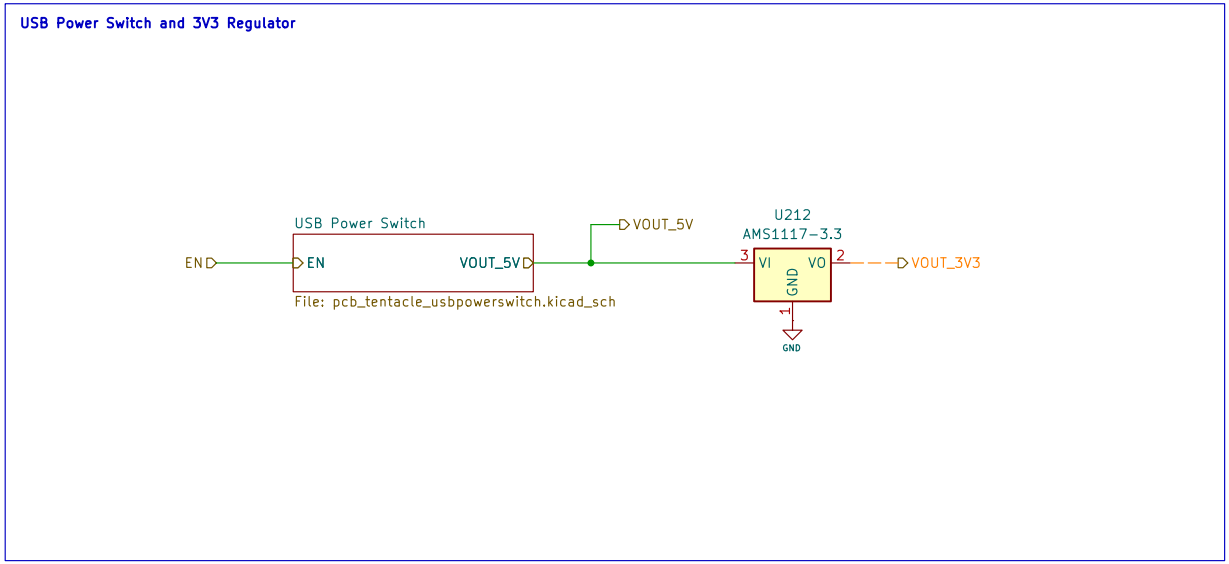


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Sheet: /Prototype Area/
File: pcb_tentacle_prototype_area.kicad_sch

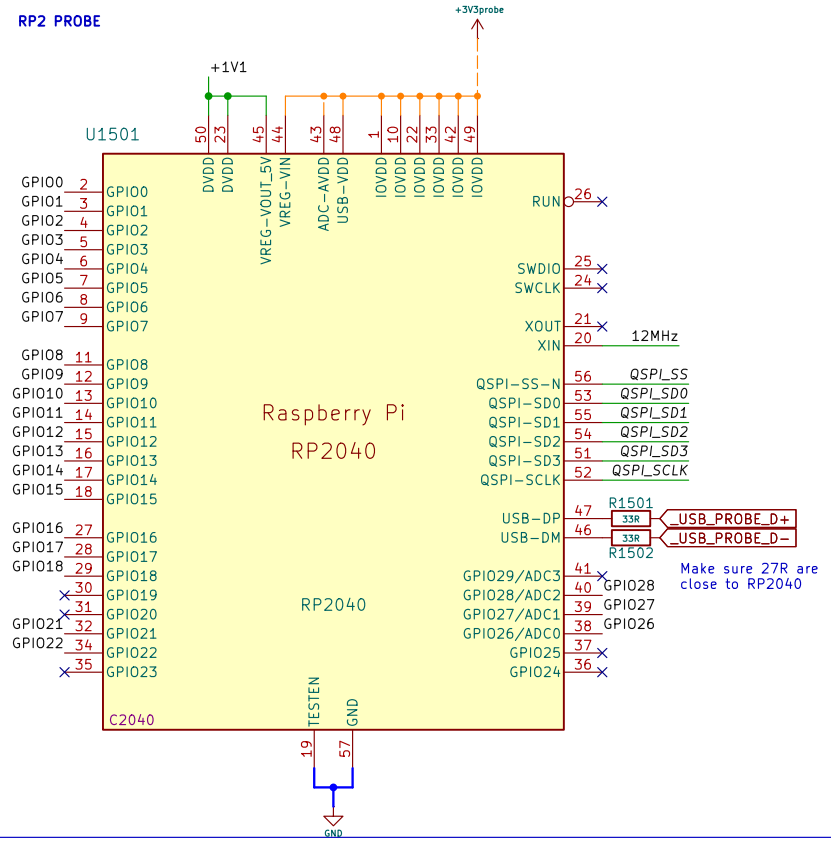
Title: Octoprobe tentacle

Size: A4	Date: 2026-10-12	**Rev: 0.9.0**
KiCad E.D.A. 10.0.4		Id: 13/21

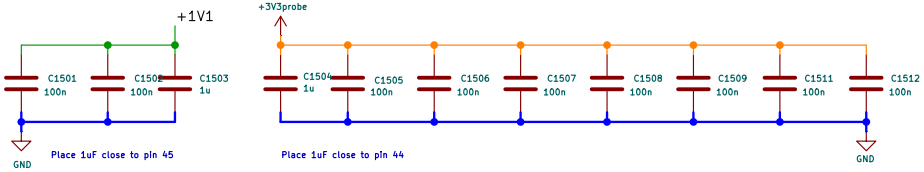


Sheet: /USB Power Switch PROBE 3V3/		
File: pcb_tentacle_usbpowerswitch3V3.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 14/21

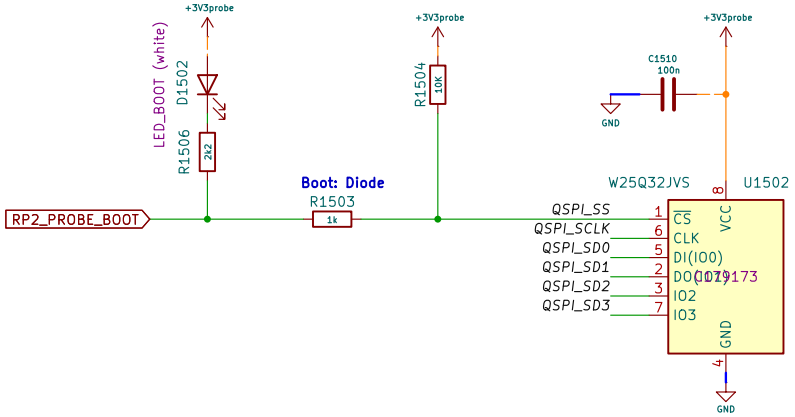
RP2 PROBE



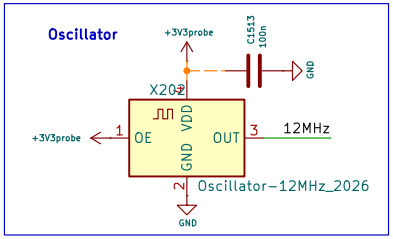
Bypass Capacitors

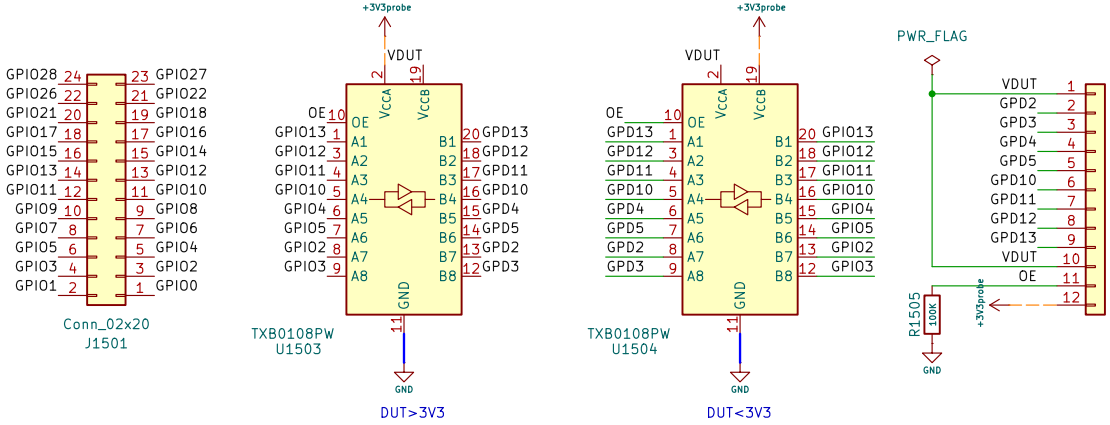


Flash Memory



Oscillator





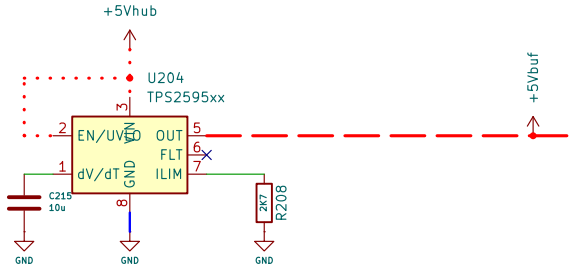
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Sheet: /RP2 PROBE/
File: pcb_tentacle_rp2_probe.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2026-10-12	**Rev: 0.9.0**
KiCad E.D.A. 10.0.4		Id: 15/21

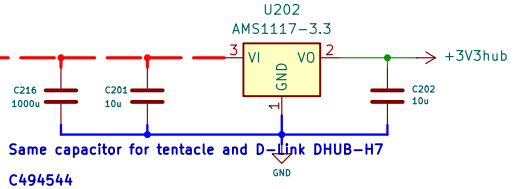
Current Limiter 5V



Datasheet "Active Current Limiting":
Goal: 800mA
Rilm = 2000/(0.8A−0.04) = R2K630, choosen: R2K7

Datasheet "Slew Rate and Inrush Current Control":
Goal: 5V/1000ms
C = 42000/V*ms = 6'363'636pF = 6uF, choosen: 10uF

Regulator 3V3

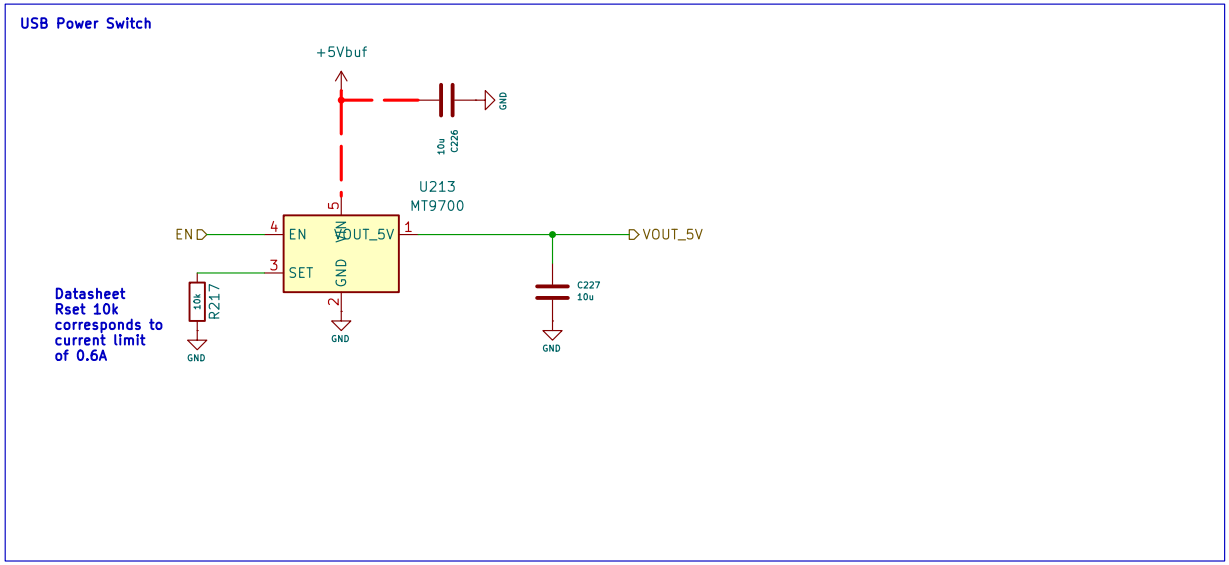


Same capacitor for tentacle and D-Link DHUB-H7

C494544

2.5mm distance between wires
6.3mm diameter
11mm height

Sheet: /Regulator USB Hub 5V, 3V3/		
File: pcb_tentacle_regulator.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 16/21

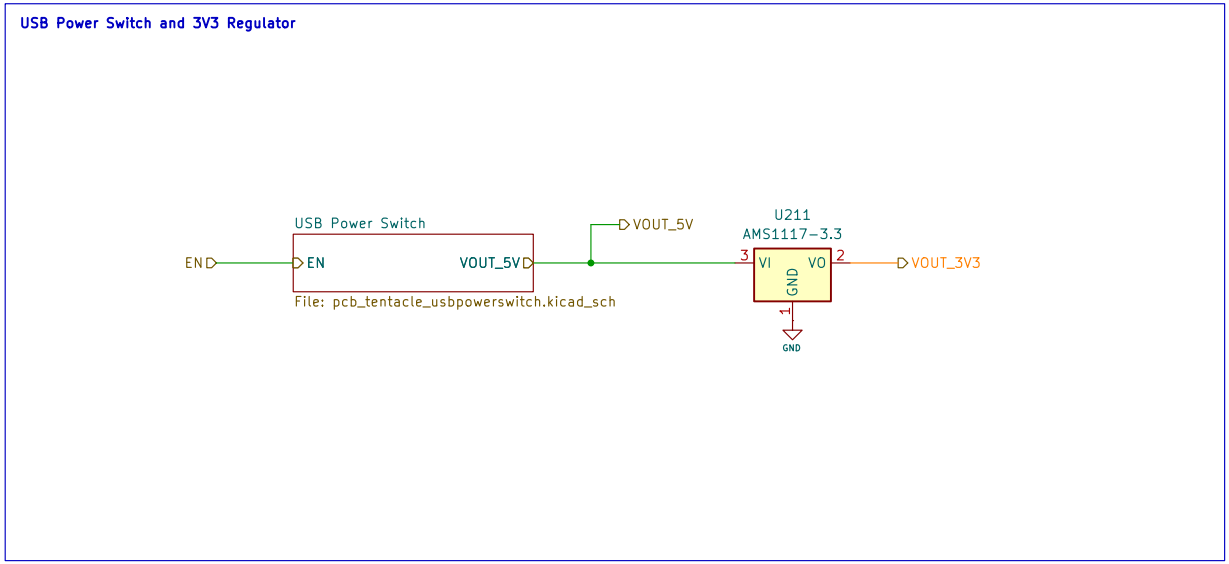


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Sheet: /USB Power Switch PROBE 3V3/USB Power Switch/
File: pcb_tentacle_usbpowerswitch.kicad_sch

Title: Octoprobe tentacle

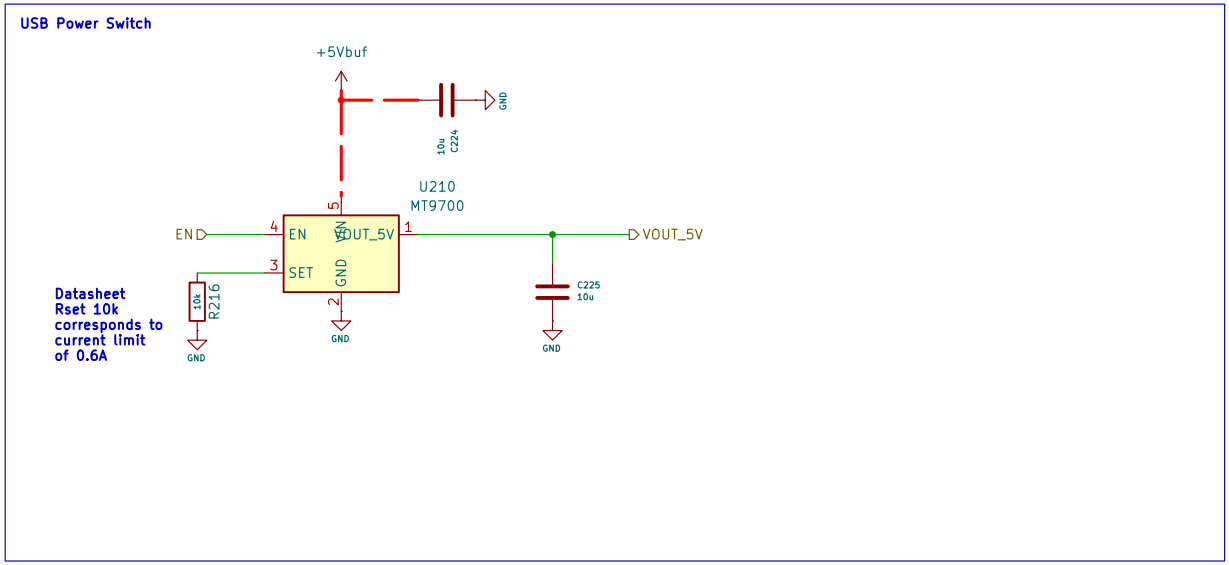
Size: A4	Date: 2025-02-13	**Rev: 0.4**
KiCad E.D.A. 10.0.4		Id: 17/21



Sheet: /USB Power Switch INFRA 3V3/
File: pcb_tentacle_usbpowerswitch3V3.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 18/21



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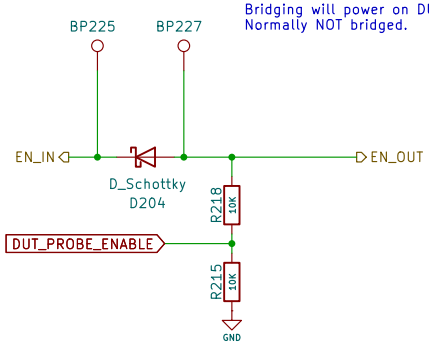
Sheet: /USB Power Switch INFRA 3V3/USB Power Switch/
File: pcb_tentacle_usbpowerswitch.kicad_sch

Title: Octoprobe tentacle

Size: A4	Date: 2025-02-13	**Rev: 0.4**
KiCad E.D.A. 10.0.4		Id: 19/21

Power enable AND

Bridging will power on DUT/RP_PROBE when plugged in.
Normally NOT bridged.



DUT_POWER_ENABLE	EN_IN	EN_OUT	Command
high impedance	0	0	After connecting tentacle
high impedance	0	0	After connecting tentacle
1	0	0	After octoprobe initialized tentacle
1	1	1	After octoprobe initialized tentacle

Sequence

Connect tentacle:
  EN_IN is 1
  EN_OUT is 0
  DUT is NOT powered
Octoprobe starts:
  * Set USB hub power off: EN_IN is 0, EN_OUT is 0
  * Set DUT_PROBE_ENABLE to VCC: EN_OUT not follows EN_IN

SITUATION WITHOUT THIS SCHEMATIC
When the tentacle is connected, linux will power on all USB ports.
This is unwanted: DUT and RP_PROBE should not be powered!

SITUATION WITH THIS SCHEMATIC
When the tentacle is connected, DUT and RP_PROBE will NOT be powered.
As soon as DUT_PROBE_ENABLE is set to VCC, USB port power will work as expected.

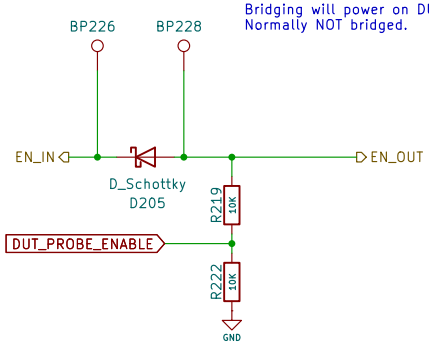
Sheet: /Power Enable AND DUT/
File: pcb_tentacle_power_enable_AND.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 20/21

Power enable AND

Bridging will power on DUT/RP_PROBE when plugged in.
Normally NOT bridged.



DUT_POWER_ENABLE	EN_IN	EN_OUT	Command
high impedance	0	0	After connecting tentacle
high impedance	0	0	After connecting tentacle
1	0	0	After octoprobe initialized tentacle
1	1	1	After octoprobe initialized tentacle

Sequence

Connect tentacle:
  EN_IN is 1
  EN_OUT is 0
  DUT is NOT powered
Octoprobe starts:
  * Set USB hub power off: EN_IN is 0, EN_OUT is 0
  * Set DUT_PROBE_ENABLE to VCC: EN_OUT not follows EN_IN

SITUATION WITHOUT THIS SCHEMATIC
When the tentacle is connected, linux will power on all USB ports.
This is unwanted: DUT and RP_PROBE should not be powered!

SITUATION WITH THIS SCHEMATIC
When the tentacle is connected, DUT and RP_PROBE will NOT be powered.
As soon as DUT_PROBE_ENABLE is set to VCC, USB port power will work as expected.

Sheet: /Power Enable AND PROBE/
File: pcb_tentacle_power_enable_AND.kicad_sch

Title:

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